

Counting Paths

Time limit: 1.00 s

Memory limit: 512 MB

You are given a tree consisting of n nodes, and m paths in the tree.

Your task is to calculate for each node the number of paths containing that node.

Input

The first input line contains integers n and m : the number of nodes and paths. The nodes are numbered $1, 2, \dots, n$.

Then there are $n - 1$ lines describing the edges. Each line contains two integers a and b : there is an edge between nodes a and b .

Finally, there are m lines describing the paths. Each line contains two integers a and b : there is a path between nodes a and b .

Output

Print n integers: for each node $1, 2, \dots, n$, the number of paths containing that node.

Constraints

- $1 \leq n, m \leq 2 \cdot 10^5$
- $1 \leq a, b \leq n$

Example

Input	Output
5 3 1 2 1 3 3 4 3 5 1 3 2 5 1 4	3 1 3 1 1